ENGR 111
Introduction to Engineering
Class #2

Which kind of engineer?

Zoom protocol

Name Set your Zoom name to your real name, or whatever you'd like me to call you.

Photo Set a professional profile photo.

Camera Encouraged, not required.

Mic Off unless you're speaking.

Your questions, anytime Type in chat or raise your hand, and I'll check periodically. Or just unmute and jump in.

My questions to the class Answer in chat, or unmute and answer.

If you speak Please state your name first.

If I drop out (e.g., my Wi-Fi fails) Stay in the meeting. I'll be back in a few minutes on my phone hotspot.

Civil engineering

The built environment: roads, bridges, buildings, transit, water, airports.

Oldest branch. Major civil projects go back more than 5,000 years.

The one that gets licensed. About 30 percent of engineers register as a P.E. For civil it is far higher, because the public safety consequences are direct.

Environmental, a separate discipline. Air pollution control, water supply, waste, public health.



Transportation



Structures



Water resources



Environmental

Space telescopes are on that list too. Still sound like roads and bridges?

Mechanical engineering

Machines: engines, turbines, robots, machine tools, heating and cooling.

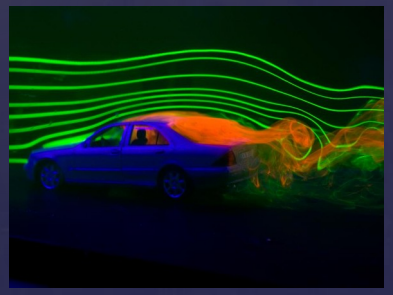
The biggest major. More students graduate in mechanical than in any other engineering discipline.

And the broadest. Machines that produce power, and machines that use it. ASME lists 35 technical divisions.

Aerospace, a separate discipline. Aircraft, missiles, spacecraft. Aerodynamics, propulsion, structures, guidance and control.



Machine design



Thermal and fluids



HVAC



Aerospace

Manufacturing and industrial

Two disciplines about making things rather than designing them. Constantly confused, genuinely separate.

Sixty percent of industrial engineers work in manufacturing. The rest turn up in hospitals, banks, government, retail, transportation, and entertainment.

Industrial is the one about people. Its professional society calls it the people-oriented engineering profession.



Process design



Tooling

Manufacturing engineering

Making the product. How the materials behave, what equipment builds it, and managing the process end to end.

Industrial engineering

Making the system work. People, machines, materials, information, and energy, in any industry at all.

Electrical engineering

Electrical devices, electrical systems, and the use of electrical energy.

Virtually every industry. Homes, cars, computers, the machines that run factories, national security systems.

IEEE has 38 technical societies. Power, signals, electronics, instrumentation, and thirty-four more.

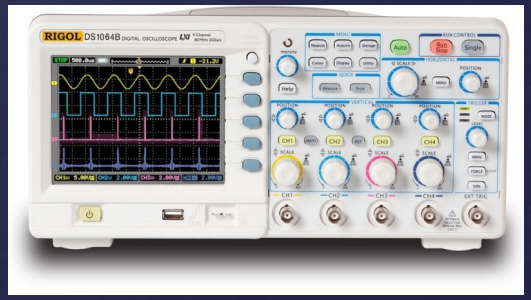
Telecommunications, a separate discipline. Modems and encoders, cellular and satellite modeling, the software behind voicemail and call waiting.



Power



Communications



Instrumentation

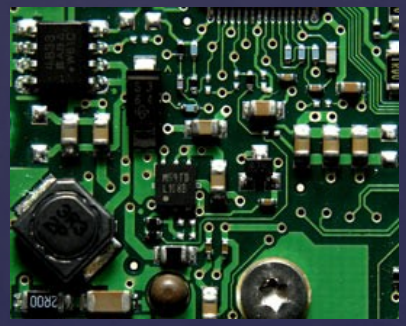
What in the room you are sitting in has no electrical engineering in it?

Computer engineering

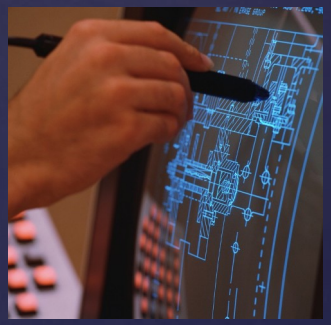
The machine itself: chips, circuit boards, networks, and the code closest to them.

It started inside electrical. First accredited program, Case Western Reserve, 1971. Still built on much of the electrical curriculum.

It is a continuum, not three boxes. No clear line between them. Plenty of people work as more than one.



Hardware



Architecture

Computer engineering

Hardware and firmware. The chips and boards themselves, and the microcode that drives them.

Software engineering

Building software as an engineering process. Requirements, architecture, quality assurance. Programming is only one phase.

Computer science

Theory and design. Creating new algorithms and new languages rather than applying existing ones.

Chemical engineering

Processes that change what matter is, at industrial scale.

Not the same as chemistry. A chemist makes it work once in a flask. A chemical engineer makes it work ten thousand times a day, profitably, without exploding.

Where you meet it. Plastics, fuels, food, drugs, synthetic fibers, shampoo, soap, cosmetics.

Biomedical, a separate discipline. Pacemakers, cochlear implants, medical imaging, laser surgery, prosthetic implants.



Refining



Food processing



Pharmaceuticals



Biomedical

What breaks when you multiply a beaker by ten thousand?

Everything else

Smaller, more specialized domains. Some are majors. Some are a concentration in somebody else's department.

Construction engineering. A cross between a civil engineer and a construction manager. The design and the site both.

Renewable energy. Wind, solar, storage, and the grid that has to absorb them. A full major at Oregon Tech.

Also on the list. Optics, for grocery scanners and eye surgery. Nuclear, for submarines and medical isotopes. Mining, ceramic, agricultural, marine, petroleum, materials.



Construction management



Wind



Solar

Did we miss any? Name one that has not been on a slide today.

Who studies it, and who actually does it

Degrees awarded, 2022

Degrees, top three. Mechanical, electrical, civil.

Jobs, top three. Industrial, civil, electrical.

Industrial is seventh in degrees, first in jobs. The category is broad. Manufacturing engineers and validation engineers are counted inside it.

Biomedical is the reverse. Sixth in degrees, last in jobs.

Software developers are on neither chart. About 1.7 million of them, counted as computer work rather than engineering.

Engineers employed, 2022

Which chart should you care about?

Sources: American Society for Engineering Education (degrees); U.S. Bureau of Labor Statistics (employment). Electrical combines electrical and electronics engineers. The nine disciplines shown total about 1.4 million jobs.

Which major is right for you?

If you choose something you like doing, work hard at it, and do it well, the money will take care of itself.

Landis, Peuker, and Mott

So: what would you like to spend your career doing?

Not what pays. Not what you think you can get into. What you would actually like to spend your days on.